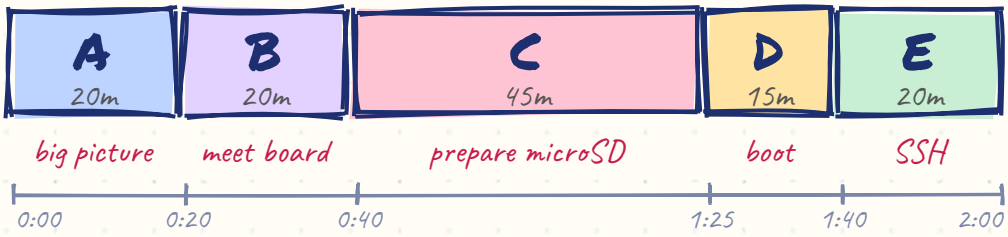
Day 1: Meet my Raspberry Pi 4!

Edge AI study course | 2 hours | Pi 4 + Windows laptop

By tonight I can...

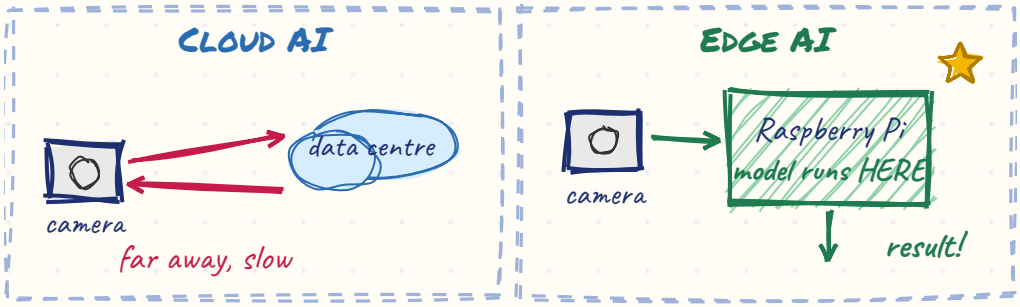
- [ ] explain Edge AI in 2 sentences
- [ ] name the main parts of the Pi 4
- [ ] write the OS onto a microSD card *(+ my own settings)*
- [ ] power the Pi, read its 2 LEDs, wait out first boot
- [ ] log in over **SSH** from Windows + shut down safely

My 2 hours



A · Big picture: what is Edge AI?

Cloud AI: my data travels to a **data centre** and back. Edge AI = the model runs ON the device next to the data (camera, robot, sensor box, a Pi).



5 reasons it matters (book)

- **Low latency** – answer in milliseconds, no round trip
- **Privacy** – raw images / audio stay on the device
- **Network independence** – works in a tunnel, field, factory
- **Less bandwidth** – send a result, not a video stream
- **Energy efficiency** – small optimised models, small power

Read: book PDF pp. 20-21 and 25-27 (printed pages are 1 lower)

My notes

questions

today I learned

TWO FAMILIES OF AI IN THIS COURSE

Fixed-function (reactive)
input -> fixed kind of answer
- image classification
- object detection
- counting with YOLO

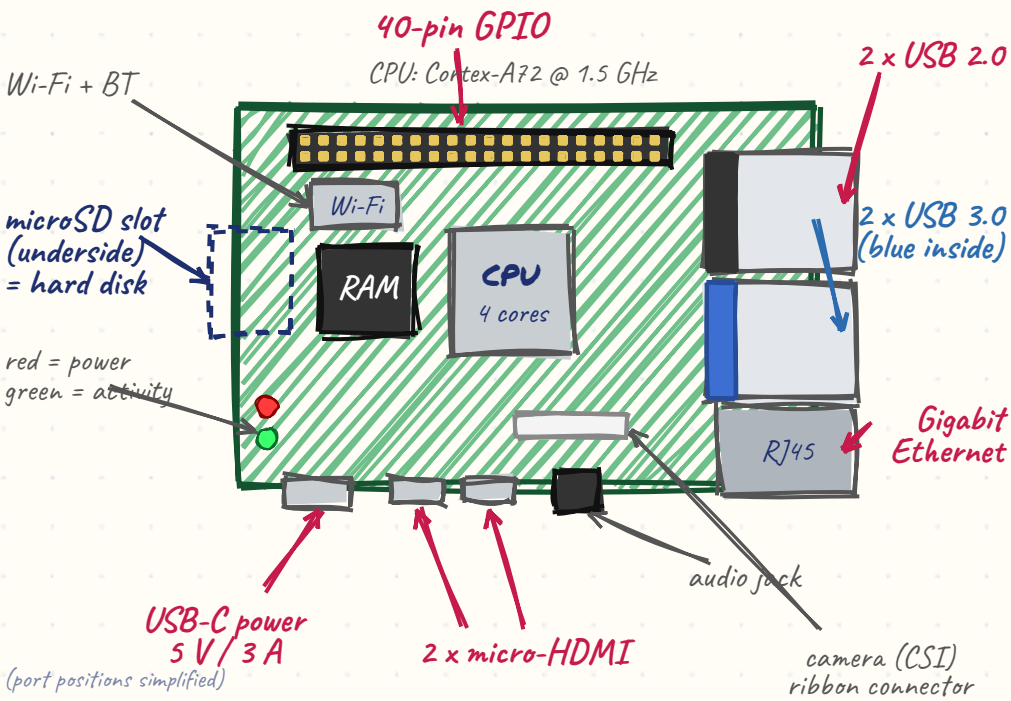
Phase 2 - days 14-26

Generative (proactive)
creates new content
- small language models
- vision-language models
- agents

Phase 3 - days 27-36

My board = Pi 4, 4 GB. Vision runs fine. For language models only the **small** ones, and slower than the book's Pi 5 numbers. RAM is the limit!

B · MEET THE BOARD



PART	WHAT IT DOES	WHY I CARE
CPU 4 cores	runs OS + code	does the neural-net maths
RAM 4 GB	working memory	models must fit in it!
microSD	the "hard disk"	always shut down first
GPIO 40 pins	sensors, LEDs	physical computing (Phase 3)
CSI / USB	camera ports	vision projects (Day 5)

My turn: find every part on my real board, take a photo, label it.
Hold the edges. No metal table. No wet fingers!

Pi 4 (mine) vs Pi 5: Pi 5 = faster CPU (2.4 vs 1.5 GHz), power button, 27 W charger, runs hotter (fan), smaller camera connectors, needs a recent OS. In Imager only the **Device** changes. SSH, commands, shutdown: identical. Restart: Pi 4 replug, Pi 5 button.

Read: book PDF pp. 29-33. The book's picture is the Pi 5.

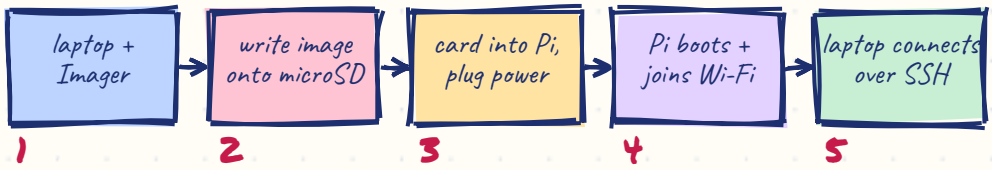
MY NOTES

my board photo notes

questions

C · Flash the microSD card (45 min)

The Pi has no hard disk. It boots the OS from the **microSD**. "Installing the OS" = writing an image onto the card with **Raspberry Pi Imager**. I bake in my settings too, so no monitor or keyboard is needed (**headless**).



settings I choose in step 1 are baked into the card

Which OS?

Raspberry Pi OS (64-bit)
with desktop – my choice

Lite = no desktop (book uses it for the tiny Zero). 32-bit: avoid.

Before I start

- [ ] Imager installed
- [ ] Wi-Fi name + password
- [ ] unplug USB drives!
- [ ] card backed up

Steps on Windows

1. Open Imager. If Windows says "format the disk?" -> **Cancel**
2. Device = `Raspberry Pi 4` (Pi 5: choose Pi 5)
3. OS = `Raspberry Pi OS (64-bit)`
4. Storage = my card (check size + drive letter!)
5. Next -> edit settings (gear icon / "customise")
6. Write -> verify -> "Write successful" -> remove card

Settings (very important)

hostname	`raspi4`
user	`piuser` + strong password
Wi-Fi	name, password, country `KW`
zone / keys	`Asia/Kuwait`, `us`
SSH	**ENABLE (password)!!**



book's Imager (v1.8.5, Mac)

Careful! Imager ERASES the card. A wrong drive = lost data. Unplug other disks.

Office Wi-Fi may need a login / block devices. Plan B: phone hotspot or Ethernet.

Read: book PDF pp. 34-37. Newer Imager = same options, wizard-style screens.

MY NOTES

my settings (not the password!)

what looked different

D · First boot (15 min)

1. Card into the slot on the **underside** (contacts toward the board)
2. Optional monitor / keyboard first
3. Power **last** (USB-C, 5 V / 3 A) - Pi 4 has **no power button**
4. Wait a few minutes. It may restart once. **Don't unplug!**

RED steady = power is good

GREEN flickers = Pi is reading the card (booting)

first boot: resize disk, apply my settings, start SSH (~1-3 min)

E · First SSH login from Windows (20 min)

SSH = Secure SHell: the Pi's command line in a window on my laptop, encrypted. Same Wi-Fi network needed!

1. Start -> **PowerShell**. Test `ssh -V`
2. Connect (below). First time: type `yes`
3. Password: **nothing shows** as I type - normal!
4. Prompt `piuser@raspi4:~ $` = I'm INSIDE the Pi



book: first login (Mac)

```
PS C:\> ssh piuser@raspi4.local
Are you sure you want to continue (yes/no)? yes
password: (invisible)
piuser@raspi4:~ $ _
```

First 5 commands

`hostname -I`	the Pi's IP address
`uname -m`	want **aarch64** (= 64-bit)
`cat /etc/os-release`	OS version - write it down!
`python3 --version`	needed on Day 4 (TFLite is picky)
`free -h`	~4 GB total

ALWAYS shut down properly: `sudo shutdown -h now`
wait until the **green LED is dark**, then unplug. Pulling the plug can corrupt the card!

Read: book PDF pp. 38-39. Slides: Class 3, slide 7 (Windows: cmd or PowerShell).

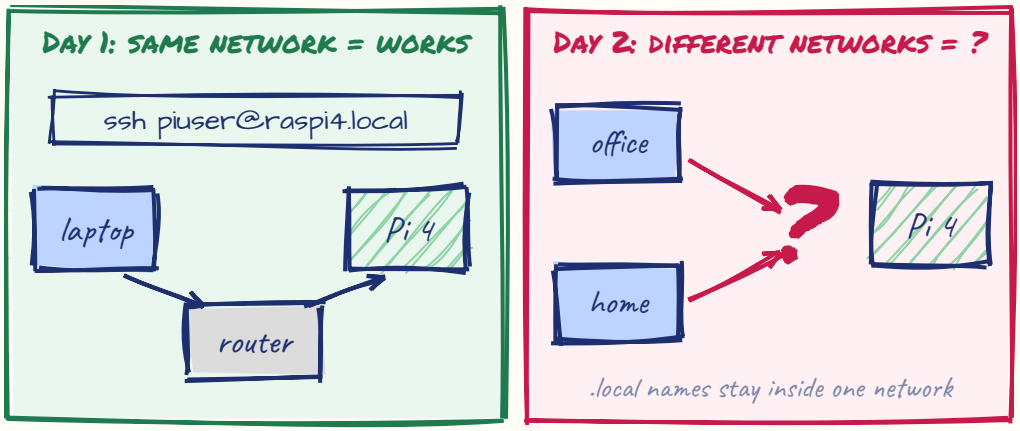
MY NOTES

my IP + OS + Python

what went wrong

Two laptops, one Pi

Office laptop + home laptop = two **windows** into the same Pi. But `raspi4.local` only works inside one network.



- **Option 1:** carry the Pi + charger, teach it both Wi-Fi networks (Day 2)
- **Option 2:** leave it in one place + remote-access service (Day 2: I'll check current options)

When it doesn't work

I see	Probably	Try
`Could not resolve hostname`	not on network yet / typo / .local blocked	wait 5 min, check Wi-Fi, use IP, Ethernet / hotspot
`Connection timed out`	different networks, VPN, guest Wi-Fi	same Wi-Fi, VPN off
`Permission denied`	wrong user / password	retype (case!) or re-flash
`HOST IDENTIFICATION HAS CHANGED`	I re-flashed the card	`ssh-keygen -R raspi4.local`
red LED, green never flickers	no OS found	reseat / re-flash / other card
no LEDs	no power	other cable (5 V / 3 A)

Quick quiz *(answer here first!)*

1. Image classification: fixed-function or generative? _____
2. Why enable SSH in Imager *before* writing? _____
3. Red LED steady = _____ Green flicker = _____
4. Unplug without shutdown -> _____
5. Why does `raspi4.local` fail from home? _____
6. Which command shows the IP? _____ 64-bit? _____
7. 3 advantages of edge AI: _____

Answers: standard tutorial, "Check yourself". Tomorrow, Day 2: remote access like a pro!

My notes

home vs office network

questions

MY CONNECTION CARD

hostname	_____
username	_____
password hint	_____
Wi-Fi 1 (home?)	_____
Wi-Fi 2 (office?)	_____
IP address (changes!)	_____
OS release	_____
Python version	_____
uname -m	_____
microSD size / brand	_____
date flashed	_____

DAY 1 DONE WHEN...

- [ ] Pi boots + red LED steady, green flickers
- [ ] PowerShell shows `piuser@raspi4:~ $`
- [ ] all 5 commands run and I wrote down OS + Python
- [ ] I shut down with `sudo shutdown -h now`
- [ ] connection card filled in

TEACH IT LATER *(my ideas)*

45-min lab: 10 min big picture, 15 flash, 10 boot + LEDs, 10 SSH + commands. Typical student mistakes: wrong drive chosen, Wi-Fi typo, SSH not enabled, guest / enterprise Wi-Fi, unplugging without shutdown, "host identification changed" after re-flash.

free space - sketches, ideas, questions for the professor...

MY NOTES

to do before Day 2

ideas